

Data wrangling fundamentals

Last time

- `filter`: choose certain rows
- `summarize`: calculate summary statistics
- `group_by`: group rows together
- `mutate`: create new columns

Input data

Key principle: The first argument to a `dplyr` function is the input dataframe

Example:

```
1 filter(starwars, species == "Droid")
```

A tibble: 2 × 14 ^{data}

	name	height	mass	hair_color	skin_color	eye_color	birth_year	sex
gender								
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<dbl>	<chr>
<chr>								
1	C-3P0	167	75	<NA>	gold	yellow	112	none
	masculine							
2	R2-D2	96	32	<NA>	white, blue	red	33	none
	masculine							

i 5 more variables: homeworld <chr>, species <chr>, films <list>,
vehicles <list>, starships <list>

Input data and the pipe

By default, the pipe (`|>`) enters the output of one line as the first argument in the next line

```
1 starwars |>
2   filter(species == "Droid")
```

Same as

`filter(starwars, species == "Droid")`

```
1 starwars |>
2   filter(species == "Droid") |>
3   summarize(mean_height = mean(height))
```

`summarize(filter(starwars, ...), mean_height = ...)`

Data in, data out

Key principle: The **input** to a dplyr function is always a data frame, and the **output** is always a dataframe

```
1 starwars |>  
2   filter(species %in% c("Droid", "Human", "Ewok")) |>  
3   group_by(species) |>  
4   summarize(mean_height = mean(height, na.rm=T))
```

← check whether species belongs to specified set

```
# A tibble: 3 × 2  
  species mean_height  
  <chr>      <dbl>  
1 Droid      131.  
2 Ewok        88  
3 Human      178
```

← data frame

The output

Question: After running this code, have we changed the original starwars data?

```
1 starwars |>
2   filter(species %in% c("Droid", "Human", "Ewok")) |>
3   group_by(species) |>
4   summarize(mean_height = mean(height, na.rm=T))
```

```
# A tibble: 3 × 2
  species mean_height
  <chr>      <dbl>
1 Droid      131.
2 Ewok        88
3 Human      178
```

The output

Key principle: `dplyr` functions do **not** modify the original data; they return a *new* data frame

```
1 starwars |>
2   filter(species %in% c("Droid", "Human", "Ewok")) |>
3   group_by(species) |>
4   summarize(mean_height = mean(height, na.rm=T))
```

```
# A tibble: 3 × 2
  species mean_height
  <chr>      <dbl>
1 Droid      131.
2 Ewok        88
3 Human      178
```

```
1 dim(starwars)
```

```
[1] 87 14
```

The output

Key principle: `dplyr` functions do **not** modify the original data; they return a *new* data frame

```
1 starwars |>
2   filter(species %in% c("Droid", "Human", "Ewok")) |>
3   group_by(species) |>
4   summarize(mean_height = mean(height, na.rm=T))
```

```
# A tibble: 3 × 2
  species mean_height
  <chr>      <dbl>
1 Droid      131.
2 Ewok        88
3 Human      178
```

Question: How do I save the output data so I can use it again?

Assignment

new object (called species_means)

assignment operator

everything on RHS get
stored in LHS

```
1 species_means <- starwars |>
2   filter(species %in% c("Droid", "Human", "Ewok")) |>
3   group_by(species) |>
4   summarize(mean_height = mean(height, na.rm=T))
5
6 species_means
```

A tibble: 3 × 2

	species	mean_height
	<chr>	<dbl>
1	Droid	131.
2	Ewok	88
3	Human	178

```
1 starwars
```

A tibble: 87 × 14

	name	height	mass	hair_color	skin_color	eye_color	birth_year	sex
	<chr>	<int>	<dbl>	<chr>	<chr>	<chr>	<dbl>	
	<chr>	<chr>						
1	Luke Sk...	172	77	blond	fair	blue	19	

Class activity

- Work with a neighbor on the class activity (handout)
- I will collect the handout at the end of class

Activity

What is the `select` function doing?

```
1 teams |>  
2   select(yearID, franchID, W, L)
```

	yearID	franchID	W	L
1	2021	BAL	52	110
2	2021	ANA	77	85
3	2021	NYM	77	85
4	2022	BAL	83	79
5	2022	ANA	73	89
6	2022	NYM	101	61
7	2023	BAL	101	61
8	2023	ANA	73	89
9	2023	NYM	75	87

choosing a subset of columns

Activity

After running the following code, what will be the dimensions (number of rows and columns) of the `team_sub` dataset? Of the `teams` dataset?

```
1 team_sub <- teams |>  
2   filter(franchID %in% c("BAL", "NYM"))
```

`team_sub`: 6 rows, 5 columns

(only looking at 2 teams)

`teams`: still 9x5 (unchanged)

Activity

After running the following code, what will be the dimensions (number of rows and columns) of the `team_sub` dataset? Of the `teams` dataset?

```
1 team_sub <- teams |>
2   filter(franchID %in% c("BAL", "NYM"))
3
4 team_sub
```

	yearID	franchID	W	L	attendance
1	2021	BAL	52	110	793229
2	2021	NYM	77	85	1511926
3	2022	BAL	83	79	1368367
4	2022	NYM	101	61	2564737
5	2023	BAL	101	61	1936798
6	2023	NYM	75	87	2573555

Activity

After running the following code, what will be the dimensions (number of rows and columns) of the `team_stats` dataset?

```
1 team_stats <- teams |>
2   group_by(franchID) |>
3   summarize(max_w = max(W), min_w = min(W))
```

· create one row of summary stats per group
→ 3 rows

· one column per summary statistic, +
column for each grouping variable (franchID)
→ 3 columns

Activity

After running the following code, what will be the dimensions (number of rows and columns) of the `team_stats` dataset?

```
1 team_stats <- teams |>
2   group_by(franchID) |>
3   summarize(max_w = max(W), min_w = min(W))
4
5 team_stats
```

```
# A tibble: 3 × 3
  franchID max_w min_w
  <fct>     <int> <int>
1 ANA         77     73
2 BAL        101     52
3 NYM        101     75
```

Activity

Give the output of the following code:

```
1 teams |>
2   filter(franchID %in% c("BAL", "NYM")) |>
3   group_by(franchID) |>
4   summarize(max_attend = max(attendance))
```

<u>franchID</u>	<u>max_attend</u>
BAL	1936798
NYM	2573555

2 rows

(column headers don't count as a row)

Activity

Give the output of the following code:

```
1 teams |>
2   filter(franchID %in% c("BAL", "NYM")) |>
3   group_by(franchID) |>
4   summarize(max_attend = max(attendance))
```

```
# A tibble: 2 × 2
  franchID max_attend
  <fct>      <int>
1 BAL         1936798
2 NYM         2573555
```

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)

Monday's class will be reserved for getting Git and GitHub setup. We will use these tools for the rest of the semester.

- Before class, work through the **Git and GitHub assignment instructions** on the course website